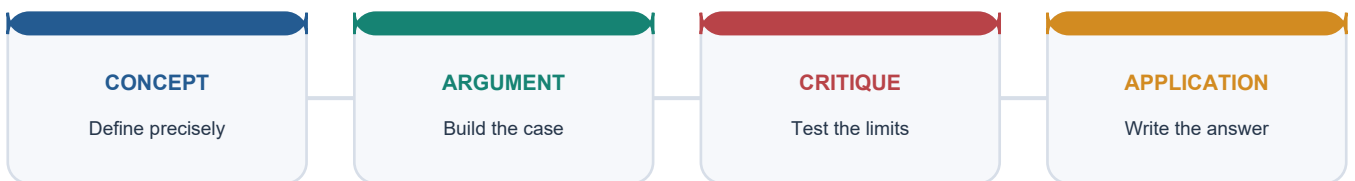


SOLVED PRACTICE WORKBOOK

Number-Systems-and-Number-Sense - CSAT Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Number-Systems-and-Number-Sense - Solved Practice Workbook 6

BASIC MCQS / REMEDIATION 6

Diagnostic set - Questions 1-16 6

Q1. Practice variant 1: What is the unit digit of 2^{11} ? 6

Q2. Practice variant 1: What is the remainder when 137 is divided by 7? 6

Q3. Practice variant 1: What is the HCF of 12 and 18? 6

Q4. Practice variant 1: How many positive divisors does 60 have? 6

Q5. Practice variant 1: How many trailing zeros are in $25!$? 6

Q6. Practice variant 1: How many multiples of 7 lie from 10 through 100, inclusive? 6

Q7. Practice variant 1: The sequence starts 3, 5, 7, What is its 8th term? 6

Q8. Practice variant 1: How many decimal digits are in 137? 6

Q9. Practice variant 2: What is the unit digit of 3^{12} ? 6

Q10. Practice variant 2: What is the remainder when 154 is divided by 8? 6

Q11. Practice variant 2: What is the HCF of 14 and 21? 6

Q12. Practice variant 2: How many positive divisors does 72 have? 6

Q13. Practice variant 2: How many trailing zeros are in $30!$? 7

Q14. Practice variant 2: How many multiples of 8 lie from 11 through 105, inclusive? 7

Q15. Practice variant 2: The sequence starts 4, 7, 10, What is its 8th term? 7

Q16. Practice variant 2: How many decimal digits are in 1037? 7

Graded set - Questions 17-32 7

Q17. Practice variant 3: What is the unit digit of 4^{13} ? 7

Q18. Practice variant 3: What is the remainder when 171 is divided by 9? 7

Q19. Practice variant 3: What is the HCF of 16 and 24? 7

Q20. Practice variant 3: How many positive divisors does 84 have? 7

Q21. Practice variant 3: How many trailing zeros are in $35!$? 7

Q22. Practice variant 3: How many multiples of 9 lie from 12 through 110, inclusive? 7

Q23. Practice variant 3: The sequence starts 5, 9, 13, What is its 8th term? 7

Q24. Practice variant 3: How many decimal digits are in 10037? 7

Q25. Practice variant 4: What is the unit digit of 5^{14} ? 7

Q26. Practice variant 4: What is the remainder when 188 is divided by 10?	7
Q27. Practice variant 4: What is the HCF of 18 and 27?	8
Q28. Practice variant 4: How many positive divisors does 96 have?	8
Q29. Practice variant 4: How many trailing zeros are in $40!$?	8
Q30. Practice variant 4: How many multiples of 10 lie from 13 through 115, inclusive?	8
Q31. Practice variant 4: The sequence starts 6, 11, 16, ... What is its 8th term?	8
Q32. Practice variant 4: How many decimal digits are in 100037?	8
Remedial and timed set - Questions 33-48	8
Q33. Practice variant 5: What is the unit digit of 6^{15} ?	8
Q34. Practice variant 5: What is the remainder when 205 is divided by 11?	8
Q35. Practice variant 5: What is the HCF of 20 and 30?	8
Q36. Practice variant 5: How many positive divisors does 108 have?	8
Q37. Practice variant 5: How many trailing zeros are in $45!$?	8
Q38. Practice variant 5: How many multiples of 11 lie from 14 through 120, inclusive?	8
Q39. Practice variant 5: The sequence starts 7, 13, 19, ... What is its 8th term?	8
Q40. Practice variant 5: How many decimal digits are in 1000037?	8
Q41. Practice variant 6: What is the unit digit of 7^{16} ?	9
Q42. Practice variant 6: What is the remainder when 222 is divided by 12?	9
Q43. Practice variant 6: What is the HCF of 22 and 33?	9
Q44. Practice variant 6: How many positive divisors does 120 have?	9
Q45. Practice variant 6: How many trailing zeros are in $50!$?	9
Q46. Practice variant 6: How many multiples of 12 lie from 15 through 125, inclusive?	9
Q47. Practice variant 6: The sequence starts 8, 15, 22, ... What is its 8th term?	9
Q48. Practice variant 6: How many decimal digits are in 10000037?	9
PYQS AND ANSWER PRACTICE	9
Complete step-by-step solutions	9
Q1. Practice variant 1: What is the unit digit of 2^{11} ?	9
Q2. Practice variant 1: What is the remainder when 137 is divided by 7?	9
Q3. Practice variant 1: What is the HCF of 12 and 18?	9
Q4. Practice variant 1: How many positive divisors does 60 have?	9
Q5. Practice variant 1: How many trailing zeros are in $25!$?	10
Q6. Practice variant 1: How many multiples of 7 lie from 10 through 100, inclusive?	10

Q7. Practice variant 1: The sequence starts 3, 5, 7, What is its 8th term?	10
Q8. Practice variant 1: How many decimal digits are in 137?	10
Q9. Practice variant 2: What is the unit digit of 3^{12} ?	10
Q10. Practice variant 2: What is the remainder when 154 is divided by 8?	10
Q11. Practice variant 2: What is the HCF of 14 and 21?	10
Q12. Practice variant 2: How many positive divisors does 72 have?	10
Q13. Practice variant 2: How many trailing zeros are in $30!$?	10
Q14. Practice variant 2: How many multiples of 8 lie from 11 through 105, inclusive?	10
Q15. Practice variant 2: The sequence starts 4, 7, 10, What is its 8th term?	11
Q16. Practice variant 2: How many decimal digits are in 1037?	11
Q17. Practice variant 3: What is the unit digit of 4^{13} ?	11
Q18. Practice variant 3: What is the remainder when 171 is divided by 9?	11
Q19. Practice variant 3: What is the HCF of 16 and 24?	11
Q20. Practice variant 3: How many positive divisors does 84 have?	11
Q21. Practice variant 3: How many trailing zeros are in $35!$?	11
Q22. Practice variant 3: How many multiples of 9 lie from 12 through 110, inclusive?	11
Q23. Practice variant 3: The sequence starts 5, 9, 13, What is its 8th term?	11
Q24. Practice variant 3: How many decimal digits are in 10037?	11
Q25. Practice variant 4: What is the unit digit of 5^{14} ?	12
Q26. Practice variant 4: What is the remainder when 188 is divided by 10?	12
Q27. Practice variant 4: What is the HCF of 18 and 27?	12
Q28. Practice variant 4: How many positive divisors does 96 have?	12
Q29. Practice variant 4: How many trailing zeros are in $40!$?	12
Q30. Practice variant 4: How many multiples of 10 lie from 13 through 115, inclusive?	12
Q31. Practice variant 4: The sequence starts 6, 11, 16, What is its 8th term?	12
Q32. Practice variant 4: How many decimal digits are in 100037?	12
Q33. Practice variant 5: What is the unit digit of 6^{15} ?	12
Q34. Practice variant 5: What is the remainder when 205 is divided by 11?	12
Q35. Practice variant 5: What is the HCF of 20 and 30?	13
Q36. Practice variant 5: How many positive divisors does 108 have?	13
Q37. Practice variant 5: How many trailing zeros are in $45!$?	13
Q38. Practice variant 5: How many multiples of 11 lie from 14 through 120, inclusive?	13

Q39. Practice variant 5: The sequence starts 7, 13, 19, What is its 8th term?	13
Q40. Practice variant 5: How many decimal digits are in 1000037?	13
Q41. Practice variant 6: What is the unit digit of 7^{16} ?	13
Q42. Practice variant 6: What is the remainder when 222 is divided by 12?	13
Q43. Practice variant 6: What is the HCF of 22 and 33?	13
Q44. Practice variant 6: How many positive divisors does 120 have?	13
Q45. Practice variant 6: How many trailing zeros are in $50!$?	14
Q46. Practice variant 6: How many multiples of 12 lie from 15 through 125, inclusive?	14
Q47. Practice variant 6: The sequence starts 8, 15, 22, What is its 8th term?	14
Q48. Practice variant 6: How many decimal digits are in 10000037?	14
Verified PYQ routing ledger	14
Final error-log checklist	16

Number-Systems-and-Number-Sense - Solved Practice Workbook

Identity: csat-02: learner-v2:g1 | Approval: false

BASIC MCQS / REMEDIATION

Diagnostic set - Questions 1-16

Q1. Practice variant 1: What is the unit digit of 2^{11} ?

A. 8 B. 9 C. 7 D. 10

Q2. Practice variant 1: What is the remainder when 137 is divided by 7?

A. 5 B. 4 C. 3 D. 6

Q3. Practice variant 1: What is the HCF of 12 and 18?

A. 7 B. 5 C. 6 D. 8

Q4. Practice variant 1: How many positive divisors does 60 have?

A. 13 B. 11 C. 14 D. 12

Q5. Practice variant 1: How many trailing zeros are in $25!$?

A. 6 B. 7 C. 5 D. 8

Q6. Practice variant 1: How many multiples of 7 lie from 10 through 100, inclusive?

A. 14 B. 13 C. 12 D. 15

Q7. Practice variant 1: The sequence starts 3, 5, 7, What is its 8th term?

A. 18 B. 16 C. 17 D. 19

Q8. Practice variant 1: How many decimal digits are in 137 ?

A. 4 B. 2 C. 5 D. 3

Q9. Practice variant 2: What is the unit digit of 3^{12} ?

A. 1 B. 2 C. 0 D. 3

Q10. Practice variant 2: What is the remainder when 154 is divided by 8?

A. 3 B. 2 C. 1 D. 4

Q11. Practice variant 2: What is the HCF of 14 and 21?

A. 8 B. 6 C. 7 D. 9

Q12. Practice variant 2: How many positive divisors does 72 have?

A. 13 B. 11 C. 14 D. 12

Q13. Practice variant 2: How many trailing zeros are in $30!$?

A. 7 B. 8 C. 6 D. 9

Q14. Practice variant 2: How many multiples of 8 lie from 11 through 105, inclusive?

A. 13 B. 12 C. 11 D. 14

Q15. Practice variant 2: The sequence starts 4, 7, 10, What is its 8th term?

A. 26 B. 24 C. 25 D. 27

Q16. Practice variant 2: How many decimal digits are in $1037?$

A. 5 B. 3 C. 6 D. 4

Graded set - Questions 17-32

Q17. Practice variant 3: What is the unit digit of 4^{13} ?

A. 4 B. 5 C. 3 D. 6

Q18. Practice variant 3: What is the remainder when 171 is divided by 9?

A. 1 B. 0 C. -1 D. 2

Q19. Practice variant 3: What is the HCF of 16 and 24?

A. 9 B. 7 C. 8 D. 10

Q20. Practice variant 3: How many positive divisors does 84 have?

A. 13 B. 11 C. 14 D. 12

Q21. Practice variant 3: How many trailing zeros are in $35!$?

A. 8 B. 9 C. 7 D. 10

Q22. Practice variant 3: How many multiples of 9 lie from 12 through 110, inclusive?

A. 12 B. 11 C. 10 D. 13

Q23. Practice variant 3: The sequence starts 5, 9, 13, What is its 8th term?

A. 34 B. 32 C. 33 D. 35

Q24. Practice variant 3: How many decimal digits are in $10037?$

A. 6 B. 4 C. 7 D. 5

Q25. Practice variant 4: What is the unit digit of 5^{14} ?

A. 5 B. 6 C. 4 D. 7

Q26. Practice variant 4: What is the remainder when 188 is divided by 10?

A. 9 B. 8 C. 7 D. 10

Q27. Practice variant 4: What is the HCF of 18 and 27?

A. 10 B. 8 C. 9 D. 11

Q28. Practice variant 4: How many positive divisors does 96 have?

A. 13 B. 11 C. 14 D. 12

Q29. Practice variant 4: How many trailing zeros are in $40!$?

A. 9 B. 10 C. 8 D. 11

Q30. Practice variant 4: How many multiples of 10 lie from 13 through 115, inclusive?

A. 11 B. 10 C. 9 D. 12

Q31. Practice variant 4: The sequence starts 6, 11, 16, What is its 8th term?

A. 42 B. 40 C. 41 D. 43

Q32. Practice variant 4: How many decimal digits are in 100037?

A. 7 B. 5 C. 8 D. 6

Remedial and timed set - Questions 33-48

Q33. Practice variant 5: What is the unit digit of 6^{15} ?

A. 6 B. 7 C. 5 D. 8

Q34. Practice variant 5: What is the remainder when 205 is divided by 11?

A. 8 B. 7 C. 6 D. 9

Q35. Practice variant 5: What is the HCF of 20 and 30?

A. 11 B. 9 C. 10 D. 12

Q36. Practice variant 5: How many positive divisors does 108 have?

A. 13 B. 11 C. 14 D. 12

Q37. Practice variant 5: How many trailing zeros are in $45!$?

A. 10 B. 11 C. 9 D. 12

Q38. Practice variant 5: How many multiples of 11 lie from 14 through 120, inclusive?

A. 10 B. 9 C. 8 D. 11

Q39. Practice variant 5: The sequence starts 7, 13, 19, What is its 8th term?

A. 50 B. 48 C. 49 D. 51

Q40. Practice variant 5: How many decimal digits are in 1000037?

A. 8 B. 6 C. 9 D. 7

Q41. Practice variant 6: What is the unit digit of 7^{16} ?

A. 1 B. 2 C. 0 D. 3

Q42. Practice variant 6: What is the remainder when 222 is divided by 12?

A. 7 B. 6 C. 5 D. 8

Q43. Practice variant 6: What is the HCF of 22 and 33?

A. 12 B. 10 C. 11 D. 13

Q44. Practice variant 6: How many positive divisors does 120 have?

A. 17 B. 15 C. 18 D. 16

Q45. Practice variant 6: How many trailing zeros are in $50!$?

A. 12 B. 13 C. 11 D. 14

Q46. Practice variant 6: How many multiples of 12 lie from 15 through 125, inclusive?

A. 10 B. 9 C. 8 D. 11

Q47. Practice variant 6: The sequence starts 8, 15, 22, What is its 8th term?

A. 58 B. 56 C. 57 D. 59

Q48. Practice variant 6: How many decimal digits are in 10000037?

A. 9 B. 7 C. 10 D. 8

PYQS AND ANSWER PRACTICE

Complete step-by-step solutions

Q1. Practice variant 1: What is the unit digit of 2^{11} ?

A. 8 B. 9 C. 7 D. 10

Correct answer: A. Reduce powers modulo 10 and use the complete unit-digit cycle.

Q2. Practice variant 1: What is the remainder when 137 is divided by 7?

A. 5 B. 4 C. 3 D. 6

Correct answer: B. Division algorithm: $n = mq + r$ with $0 \leq r < m$.

Q3. Practice variant 1: What is the HCF of 12 and 18?

A. 7 B. 5 C. 6 D. 8

Correct answer: C. Prime factorisation or Euclid's algorithm gives the greatest common divisor.

Q4. Practice variant 1: How many positive divisors does 60 have?

A. 13 B. 11 C. 14 D. 12

Correct answer: D. If $n = \text{product } p_i^{a_i}$, the divisor count is $\text{product } (a_i + 1)$.

Q5. Practice variant 1: How many trailing zeros are in 25! ?

A. 6 B. 7 C. 5 D. 8

Correct answer: A. Count factors of 5 by repeated division; factors of 2 are more abundant.

Q6. Practice variant 1: How many multiples of 7 lie from 10 through 100, inclusive?

A. 14 B. 13 C. 12 D. 15

Correct answer: B. Use $\text{floor}(\text{high}/d) - \text{floor}((\text{low}-1)/d)$.

Q7. Practice variant 1: The sequence starts 3, 5, 7, What is its 8th term?

A. 18 B. 16 C. 17 D. 19

Correct answer: C. This is an arithmetic progression: $a_n = a + (n-1)d$.

Q8. Practice variant 1: How many decimal digits are in 137?

A. 4 B. 2 C. 5 D. 3

Correct answer: D. A positive integer n has $\text{floor}(\log_{10} n) + 1$ digits; direct place-value bounding gives the same answer.

Q9. Practice variant 2: What is the unit digit of 3^{12} ?

A. 1 B. 2 C. 0 D. 3

Correct answer: A. Reduce powers modulo 10 and use the complete unit-digit cycle.

Q10. Practice variant 2: What is the remainder when 154 is divided by 8?

A. 3 B. 2 C. 1 D. 4

Correct answer: B. Division algorithm: $n = mq + r$ with $0 \leq r < m$.

Q11. Practice variant 2: What is the HCF of 14 and 21?

A. 8 B. 6 C. 7 D. 9

Correct answer: C. Prime factorisation or Euclid's algorithm gives the greatest common divisor.

Q12. Practice variant 2: How many positive divisors does 72 have?

A. 13 B. 11 C. 14 D. 12

Correct answer: D. If $n = \text{product } p_i^{a_i}$, the divisor count is $\text{product } (a_i + 1)$.

Q13. Practice variant 2: How many trailing zeros are in 30! ?

A. 7 B. 8 C. 6 D. 9

Correct answer: A. Count factors of 5 by repeated division; factors of 2 are more abundant.

Q14. Practice variant 2: How many multiples of 8 lie from 11 through 105, inclusive?

A. 13 B. 12 C. 11 D. 14

Correct answer: B. Use $\text{floor}(\text{high}/d) - \text{floor}((\text{low}-1)/d)$.

Q15. Practice variant 2: The sequence starts 4, 7, 10, What is its 8th term?

A. 26 B. 24 C. 25 D. 27

Correct answer: C. This is an arithmetic progression: $a_n = a + (n-1)d$.

Q16. Practice variant 2: How many decimal digits are in 1037?

A. 5 B. 3 C. 6 D. 4

Correct answer: D. A positive integer n has $\text{floor}(\log_{10} n) + 1$ digits; direct place-value bounding gives the same answer.

Q17. Practice variant 3: What is the unit digit of 4^{13} ?

A. 4 B. 5 C. 3 D. 6

Correct answer: A. Reduce powers modulo 10 and use the complete unit-digit cycle.

Q18. Practice variant 3: What is the remainder when 171 is divided by 9?

A. 1 B. 0 C. -1 D. 2

Correct answer: B. Division algorithm: $n = mq + r$ with $0 \leq r < m$.

Q19. Practice variant 3: What is the HCF of 16 and 24?

A. 9 B. 7 C. 8 D. 10

Correct answer: C. Prime factorisation or Euclid's algorithm gives the greatest common divisor.

Q20. Practice variant 3: How many positive divisors does 84 have?

A. 13 B. 11 C. 14 D. 12

Correct answer: D. If $n = \text{product } p_i^{a_i}$, the divisor count is $\text{product } (a_i + 1)$.

Q21. Practice variant 3: How many trailing zeros are in $35!$?

A. 8 B. 9 C. 7 D. 10

Correct answer: A. Count factors of 5 by repeated division; factors of 2 are more abundant.

Q22. Practice variant 3: How many multiples of 9 lie from 12 through 110, inclusive?

A. 12 B. 11 C. 10 D. 13

Correct answer: B. Use $\text{floor}(\text{high}/d) - \text{floor}((\text{low}-1)/d)$.

Q23. Practice variant 3: The sequence starts 5, 9, 13, What is its 8th term?

A. 34 B. 32 C. 33 D. 35

Correct answer: C. This is an arithmetic progression: $a_n = a + (n-1)d$.

Q24. Practice variant 3: How many decimal digits are in 10037?

A. 6 B. 4 C. 7 D. 5

Correct answer: D. A positive integer n has $\text{floor}(\log_{10} n) + 1$ digits; direct place-value bounding gives the same answer.

Q25. Practice variant 4: What is the unit digit of 5^{14} ?

A. 5 B. 6 C. 4 D. 7

Correct answer: A. Reduce powers modulo 10 and use the complete unit-digit cycle.

Q26. Practice variant 4: What is the remainder when 188 is divided by 10?

A. 9 B. 8 C. 7 D. 10

Correct answer: B. Division algorithm: $n = mq + r$ with $0 \leq r < m$.

Q27. Practice variant 4: What is the HCF of 18 and 27?

A. 10 B. 8 C. 9 D. 11

Correct answer: C. Prime factorisation or Euclid's algorithm gives the greatest common divisor.

Q28. Practice variant 4: How many positive divisors does 96 have?

A. 13 B. 11 C. 14 D. 12

Correct answer: D. If $n = \text{product } p_i^{a_i}$, the divisor count is $\text{product } (a_i + 1)$.

Q29. Practice variant 4: How many trailing zeros are in $40!$?

A. 9 B. 10 C. 8 D. 11

Correct answer: A. Count factors of 5 by repeated division; factors of 2 are more abundant.

Q30. Practice variant 4: How many multiples of 10 lie from 13 through 115, inclusive?

A. 11 B. 10 C. 9 D. 12

Correct answer: B. Use $\text{floor}(\text{high}/d) - \text{floor}((\text{low}-1)/d)$.

Q31. Practice variant 4: The sequence starts 6, 11, 16, What is its 8th term?

A. 42 B. 40 C. 41 D. 43

Correct answer: C. This is an arithmetic progression: $a_n = a + (n-1)d$.

Q32. Practice variant 4: How many decimal digits are in 100037?

A. 7 B. 5 C. 8 D. 6

Correct answer: D. A positive integer n has $\text{floor}(\log_{10} n) + 1$ digits; direct place-value bounding gives the same answer.

Q33. Practice variant 5: What is the unit digit of 6^{15} ?

A. 6 B. 7 C. 5 D. 8

Correct answer: A. Reduce powers modulo 10 and use the complete unit-digit cycle.

Q34. Practice variant 5: What is the remainder when 205 is divided by 11?

A. 8 B. 7 C. 6 D. 9

Correct answer: B. Division algorithm: $n = mq + r$ with $0 \leq r < m$.

Q35. Practice variant 5: What is the HCF of 20 and 30?

A. 11 B. 9 C. 10 D. 12

Correct answer: C. Prime factorisation or Euclid's algorithm gives the greatest common divisor.

Q36. Practice variant 5: How many positive divisors does 108 have?

A. 13 B. 11 C. 14 D. 12

Correct answer: D. If $n = \text{product } p_i^{a_i}$, the divisor count is $\text{product } (a_i + 1)$.

Q37. Practice variant 5: How many trailing zeros are in $45!$?

A. 10 B. 11 C. 9 D. 12

Correct answer: A. Count factors of 5 by repeated division; factors of 2 are more abundant.

Q38. Practice variant 5: How many multiples of 11 lie from 14 through 120, inclusive?

A. 10 B. 9 C. 8 D. 11

Correct answer: B. Use $\text{floor}(\text{high}/d) - \text{floor}((\text{low}-1)/d)$.

Q39. Practice variant 5: The sequence starts 7, 13, 19, What is its 8th term?

A. 50 B. 48 C. 49 D. 51

Correct answer: C. This is an arithmetic progression: $a_n = a + (n-1)d$.

Q40. Practice variant 5: How many decimal digits are in 1000037?

A. 8 B. 6 C. 9 D. 7

Correct answer: D. A positive integer n has $\text{floor}(\log_{10} n) + 1$ digits; direct place-value bounding gives the same answer.

Q41. Practice variant 6: What is the unit digit of 7^{16} ?

A. 1 B. 2 C. 0 D. 3

Correct answer: A. Reduce powers modulo 10 and use the complete unit-digit cycle.

Q42. Practice variant 6: What is the remainder when 222 is divided by 12?

A. 7 B. 6 C. 5 D. 8

Correct answer: B. Division algorithm: $n = mq + r$ with $0 \leq r < m$.

Q43. Practice variant 6: What is the HCF of 22 and 33?

A. 12 B. 10 C. 11 D. 13

Correct answer: C. Prime factorisation or Euclid's algorithm gives the greatest common divisor.

Q44. Practice variant 6: How many positive divisors does 120 have?

A. 17 B. 15 C. 18 D. 16

Correct answer: D. If $n = \text{product } p_i^{a_i}$, the divisor count is $\text{product } (a_i + 1)$.

Q45. Practice variant 6: How many trailing zeros are in 50! ?

A. 12 B. 13 C. 11 D. 14

Correct answer: A. Count factors of 5 by repeated division; factors of 2 are more abundant.**Q46. Practice variant 6: How many multiples of 12 lie from 15 through 125, inclusive?**

A. 10 B. 9 C. 8 D. 11

Correct answer: B. Use $\text{floor}(\text{high}/d) - \text{floor}((\text{low}-1)/d)$.**Q47. Practice variant 6: The sequence starts 8, 15, 22, What is its 8th term?**

A. 58 B. 56 C. 57 D. 59

Correct answer: C. This is an arithmetic progression: $a_n = a + (n-1)d$.**Q48. Practice variant 6: How many decimal digits are in 10000037?**

A. 9 B. 7 C. 10 D. 8

Correct answer: D. A positive integer n has $\text{floor}(\log_{10} n) + 1$ digits; direct place-value bounding gives the same answer.**Verified PYQ routing ledger**

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2024	6	Operator-placement minimum	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	9	Trailing-zero count	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	10	Cumulative savings sequence	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	15	Divisibility test	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	17	Terminal nonzero digit	D (supplied)	Use place value, cyclicity or a constrained enumeration.
2024	18	Common-divisor remainder	C (supplied)	Translate to congruences and plug the result into every condition.
2024	19	HCF container sizing	B (supplied)	Use prime factors or Euclid; verify the common-divisor condition.
2024	20	Prime-sum unit digit	C (supplied)	Use place value, cyclicity or a constrained enumeration.
2024	39	Expression parity	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	46	Periodic-sequence sum	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	54	Exponential divisibility	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2024	55	Integer-pair uniqueness	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	79	Number series	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	5	Factor-sum enumeration	C (supplied)	Prime-factorise, count exponents and verify divisibility.
2025	6	Prime-progression sums	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	7	Unit-fraction solution count	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	16	Derived-operator pattern	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	18	Simultaneous remainder constraints	C (supplied)	Translate to congruences and plug the result into every condition.
2025	20	Odd-product unit digit	C (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	25	Divisibility-exclusion count	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	27	Prime-expression count	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	28	LCM-HCF triple count	D (supplied)	Use prime factors or Euclid; verify the common-divisor condition.
2025	35	Highest factor power	B (supplied)	Prime-factorise, count exponents and verify divisibility.
2025	36	Number-series completion	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	40	Square-root digit count	B (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	46	Power-sum remainder	C (supplied)	Translate to congruences and plug the result into every condition.
2025	47	Repdigit HCF	C (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	48	Concatenated-sequence digit	D (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	60	Number-series completion	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	64	Factor-count invariance	C (supplied)	Prime-factorise, count exponents and verify divisibility.
2025	67	Power-series remainder	A (supplied)	Translate to congruences and plug the result into every condition.

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2025	80	Interior-multiples count	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	10	LCM multiple count in range	D (provisional)	Use prime powers and range bounds.
2026	18	String-pattern position	D (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	19	Exponent equation factorisation	B (provisional)	Prime-factorise, count exponents and verify divisibility.
2026	22	Measurement-combination optimisation	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	29	Discrete-set expression extrema	C (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	42	Unit digit of power product	C (provisional)	Use place value, cyclicity or a constrained enumeration.
2026	45	Count of powers of two	C (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	49	Digit count in two-digit integers	B (provisional)	Use place value, cyclicity or a constrained enumeration.
2026	57	Fibonacci-like recurrence	D (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	59	HCF-LCM cube relation	C (provisional)	Use prime factors or Euclid; verify the common-divisor condition.
2026	60	Divisibility-by-eleven digit puzzle	D (provisional)	Use place value, cyclicity or a constrained enumeration.
2026	66	Reversed-digit divisibility count	B (provisional)	Use place value, cyclicity or a constrained enumeration.

Final error-log checklist

- Was the family classified correctly?
- Were units, domains, quantifiers and constraints copied correctly?
- Was the first-principles relation written before a shortcut?
- Was the selected option independently recomputed or supported by the passage?
- Was the mistake logged and followed by a fresh remedial item?